

The Normal Distribution

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Probability distributions ([Note 240](#)); Probability density function (PDF) ([Note 242](#)).
- **Leads to:** Z-score outlier method ([Note 251](#)); Standard normal and the z-table ([Note 251](#)); Q-Q plot ([Note 260](#)); Central limit theorem ([Note 271](#)).
- **Compare with:** Student's t-distribution ([Note 282](#)).

1. Overview

Key point: The normal distribution is the symmetric, bell-shaped continuous distribution set by two parameters, the mean μ and the standard deviation σ ; most values sit near the mean and fewer and fewer lie far from it.

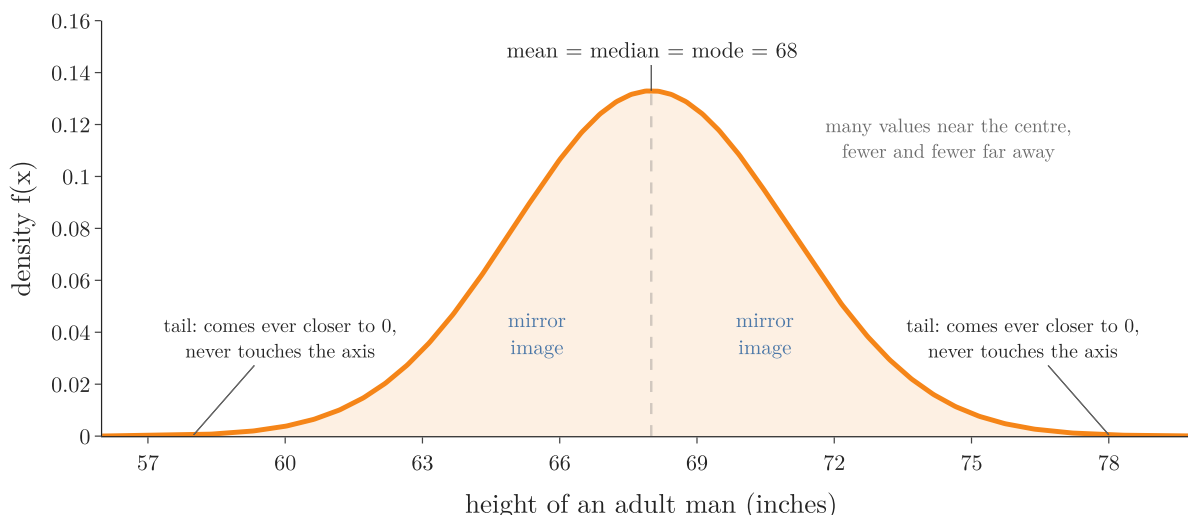


Figure 1: The normal curve of adult men's heights, $N(68, 3)$ inches

Figure 1 shows the most famous probability distribution: the heights of adult men in a population, with mean 68 inches and standard deviation 3 inches. The normal distribution was met briefly in earlier Notes: its 68-95-99.7 rule in the [z-score outliers Note](#), its formula in the [Gaussian Naive Bayes Note](#), its parameters in the [random variables and distributions Note](#).

This Note studies it properly: what it is, why it matters, where its formula comes from, its properties and its CDF. The [standard normal distribution Note](#) then turns it into probabilities with the z-table, and the [skewness Note](#) measures how far real data departs from its symmetry.

2. What the normal distribution is

Key point: A continuous distribution, symmetric around its mean, with a bell-shaped PDF whose tails never quite reach zero.

The **normal distribution**, also called the **Gaussian distribution** or the bell curve, is a continuous probability distribution that is commonly used in statistical analysis. Its PDF is symmetric around the mean and shaped like a bell. Figure 1 points out its parts:

- **The centre:** the peak sits at the mean, here 68 inches.
- **The tails:** the two ends where the curve flattens out. They come closer and closer to the x axis without ever touching it: the curve is **asymptotic** to the axis. Any value, however extreme, has some tiny density.
- **The shape:** many values lie near the centre, some lie far below it and some far above it, fewer and fewer the further out we go.

The shape point is the whole summary of the normal distribution. The y axis is probability density (see the [PDF and continuous CDF Note](#)): high near 68 because many men have heights around there, low at 59 or 77 because few do.

3. Parameters and notation

Key point: Two numbers describe a normal distribution completely: the mean μ (its centre) and the standard deviation σ (its spread); we write $X \sim N(\mu, \sigma^2)$.

A normal distribution has two parameters:

- μ , **the mean:** the centre of the distribution. Changing it slides the curve left or right without changing its shape.
- σ , **the standard deviation:** the spread. A larger σ makes the curve wider and lower; a smaller one makes it narrower and taller (the area stays 1).

Figure 5 of the [random variables and distributions Note](#) shows both effects, and the Notebook (`notebook.ipynb`) has two sliders to try them.

In books the normal distribution is written as

$$X \sim N(\mu, \sigma^2)$$

read “ X follows a normal distribution with mean μ and variance σ^2 ”. The heights of Figure 1 are $X \sim N(68, 3^2)$. Some books write $N(\mu, \sigma)$ instead: the same curve, written with the standard deviation.

So once we know that a variable is normal, its mean and standard deviation are all we need to draw its exact curve and compute any probability about it.

4. Why the normal distribution matters

Key point: The normal shape appears again and again in nature and in man-made data, and it has been studied so thoroughly that everything about it is known.

The name says it: the shape is **normal**, meaning common. Many natural phenomena follow it, for example:

- heights of people;
- weights of objects of the same kind;
- IQ scores of a population;
- measurement errors in repeated measurements (Taylor 1997, ch. 5).

For many years, people in many fields collected data and drew its PDF, and this same bell kept appearing. So it was studied in great depth, and today its mathematics is completely worked out. Because of this complete theory, given new data, we are pleased when it turns out to be roughly normal: everything known about the normal distribution then applies to it. Its uses in data science are listed in the [standard normal distribution Note](#).

5. The PDF of the normal distribution

Key point: The normal PDF is a formula in x with only two parameters, μ and σ ; plug them in and it gives the density at any value.

The bell is a graph, so it has an equation $y = f(x)$, where y is the probability density at the value x .

1. **In words:** measure how far x is from the mean in units of standard deviation, square it, halve it, and take e to minus that; then divide by $\sigma\sqrt{2\pi}$.
2. **Formula:**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Here $\pi = 3.1416$ and $e = 2.7183$ are constants, x is the variable, and μ and σ are the only parameters.

3. **Example:** heights with $\mu = 68$ and $\sigma = 3$. At the mean, $x = 68$, the exponent is 0 and $e^0 = 1$:

$$f(68) = \frac{1}{3\sqrt{2\pi}} = \frac{1}{7.520} = 0.1330$$

At $x = 72$, the distance is $4/3 = 1.333$ standard deviations:

$$f(72) = 0.1330 \times e^{-\frac{1}{2}(1.333)^2} = 0.1330 \times e^{-0.889} = 0.1330 \times 0.411 = 0.0547$$

Python: The normal PDF in scipy.

```
from scipy import stats
```

```
heights = stats.norm(loc=68, scale=3) # mu, sigma
heights.pdf(68) # 0.1330
```

5.1 Where the formula comes from

Key point: Start from the bell e^{-x^2} ; subtract μ to move it, divide the exponent by $2\sigma^2$ to widen it, and divide by $\sigma\sqrt{2\pi}$ to make its area 1.

The formula looks frightening, but it can be built term by term (Figure 2):

1. $y = e^x$ is exponential growth: it climbs faster and faster.
2. $y = e^{-x}$ puts a minus sign in front: exponential decay.
3. $y = e^{-x^2}$ squares x . Now $x = 2$ and $x = -2$ give the same value, so the curve decays on **both** sides of 0: a bell. The bell e^{-x^2} is the heart of the normal distribution. However far x moves from 0, in either direction, y gets smaller but never reaches 0.
4. $y = e^{-(x-\mu)^2}$ moves the bell so that its peak sits at μ instead of 0. The shift lets the formula describe a normal distribution centred anywhere: at 68 inches, at -2 , anywhere.
5. **Divide the exponent by $2\sigma^2$** , giving $y = e^{-(x-\mu)^2/(2\sigma^2)}$. The division stretches the bell sideways: a larger σ , a wider bell.
6. **Divide by $\sigma\sqrt{2\pi}$** . The area under every PDF must be 1. The area under the curve of step 5 turns out to be exactly $\sigma\sqrt{2\pi}$ (3.76 for $\sigma = 1.5$), so dividing by it brings the area to 1.

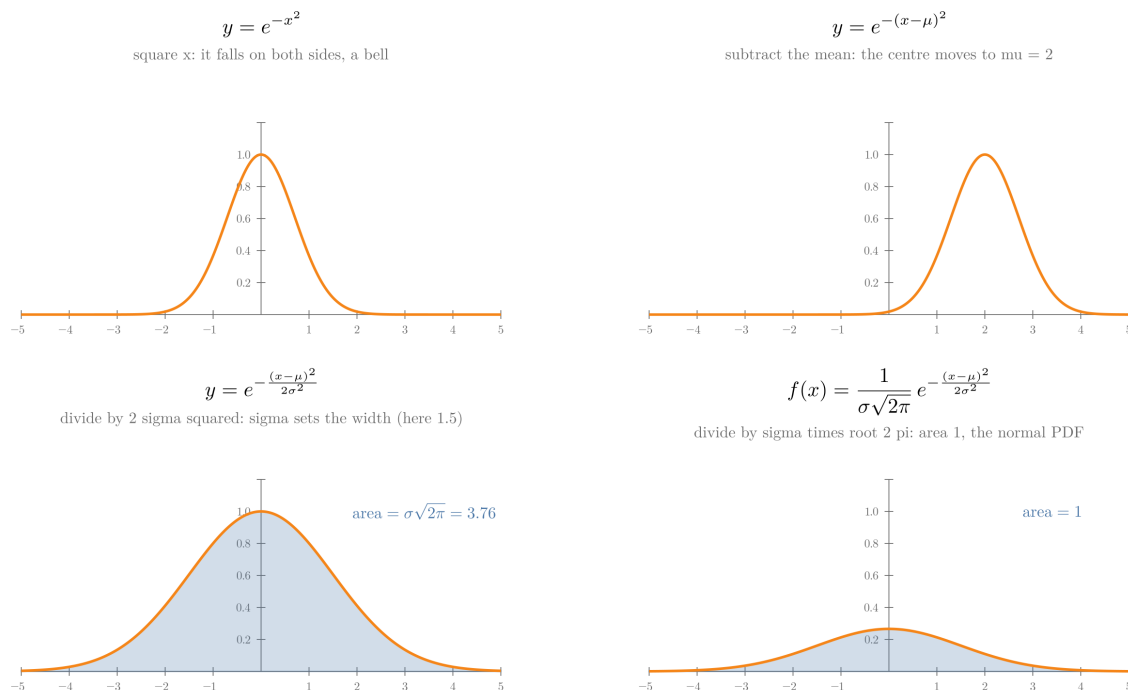


Figure 2: Building the normal PDF term by term: the bell, the shift, the width, the area

Extra: Why $2\sigma^2$ and not just σ^2 ? With the 2, the parameter σ comes out as exactly the standard deviation of the curve (MML §6.5), and the bell's two **inflection points** (where it switches from bending down to bending up) sit exactly at $\mu - \sigma$ and $\mu + \sigma$. The inflection points follow from two derivatives of f :

$$f'(x) = -\frac{x - \mu}{\sigma^2} f(x), \quad f''(x) = \frac{f(x)}{\sigma^2} \left[\frac{(x - \mu)^2}{\sigma^2} - 1 \right]$$

f'' changes sign where $(x - \mu)^2 = \sigma^2$, that is at $x = \mu \pm \sigma$. Without the 2 the same steps would give $\mu \pm \sigma/\sqrt{2}$.

The area $\sigma\sqrt{2\pi}$ in step 6 comes from a classic integral, $\int_{-\infty}^{\infty} e^{-t^2/2} dt = \sqrt{2\pi} = 2.5066$; we do not prove it here, but the Notebook checks it numerically.

6. Properties of the normal distribution

Key point: Symmetric around the mean; mean, median and mode are equal; 68%, 95% and 99.7% of values lie within 1, 2 and 3 standard deviations; total area 1.

Four properties make the normal distribution so convenient:

1. **Symmetry.** The curve is symmetric around the mean: the left half is a mirror image of the right half (Figure 1). So a probability computed on one side gives the matching probability on the other side for free.
2. **Mean = median = mode.** For a true normal distribution the three measures of central tendency (see the [measures of central tendency Note](#)) are exactly equal: the peak (mode) is also the middle value (median) and the average (mean).
3. **The empirical rule.** About 68% of values lie within 1 standard deviation of the mean, 95% within 2 and 99.7% within 3 (see the [z-score outliers Note](#)). The [standard normal distribution Note](#) derives these numbers from the z-table.
4. **Total area 1.** The area under the curve is exactly 1, as for every PDF.

7. The CDF of the normal distribution

Key point: The normal CDF is an S-shaped curve that passes 0.5 at the mean; the smaller σ , the steeper the S.

Every PDF has a CDF, $F(x) = P(X \leq x)$: the area under the PDF from the far left up to x (see the [PDF and continuous CDF Note](#)). Figure 3 shows the PDFs of four normal distributions (top) and their CDFs (bottom).

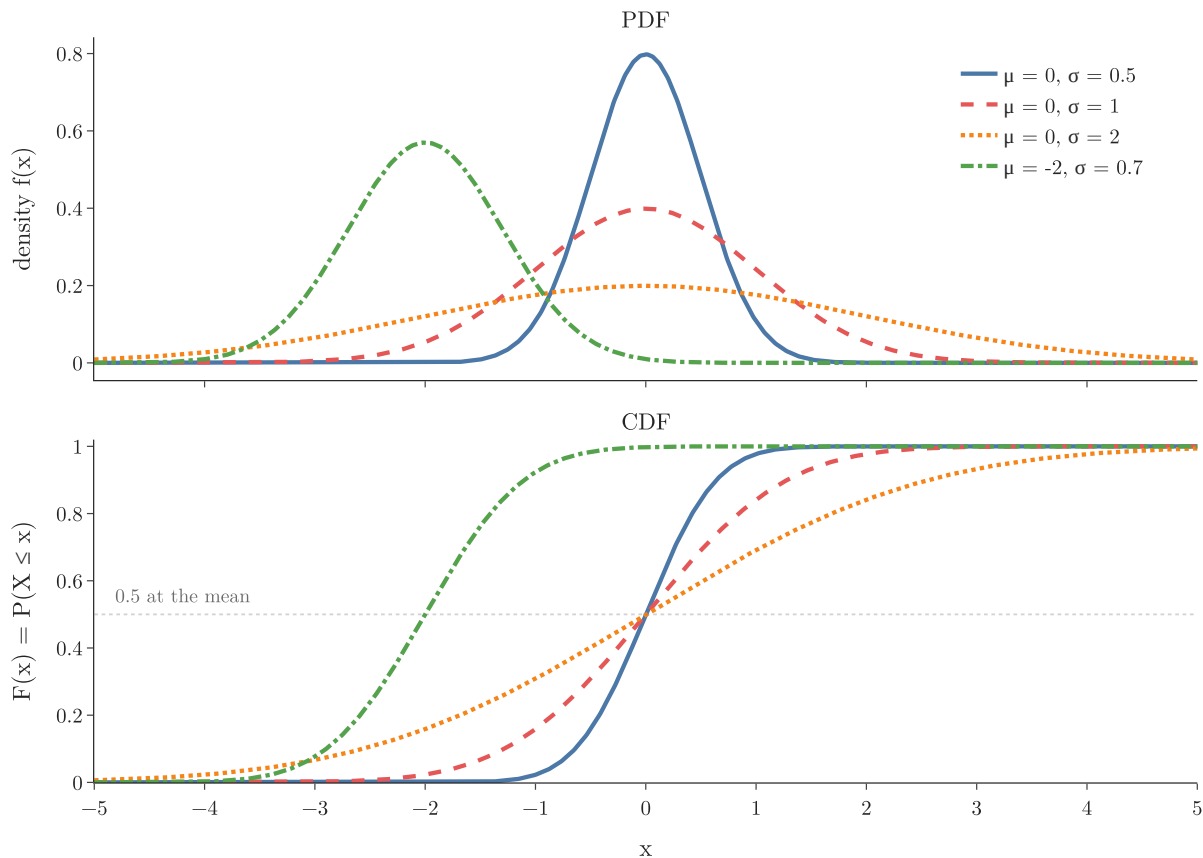


Figure 3: Normal PDFs (top) and their CDFs (bottom)

Reading Figure 3:

- **Every CDF passes 0.5 at its mean.** By symmetry, half the area lies left of the mean. For the green curve, with $\mu = -2$, $P(X \leq -2) = 0.5$; for the heights, $P(X \leq 68) = 0.5$.
- **The standard deviation sets the steepness.** With $\sigma = 0.5$ (blue) the S rises sharply near 0; with $\sigma = 2$ (orange) it rises slowly and reaches 1 far from the centre.
- **The shape is an S.** The S looks like the sigmoid function of the [sigmoid function Note](#), though the two are different functions.

1. **In words:** the CDF at x is the area under the normal PDF from minus infinity up to x .

2. **Formula:**

$$F(x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

3. **Example:** for the heights, the area up to 72 inches is

$$F(72) = P(X \leq 72) = 0.909$$

so about 91% of men are 72 inches or shorter.

Extra: The normal CDF integral has no formula in elementary functions: no combination of powers, logs and exponentials gives the normal CDF exactly (Rosenlicht 1972). So the normal CDF is computed numerically, either by software or, in the past, by printed tables. Those tables are the z-tables of the [standard normal distribution Note](#).

Python: The CDF in scipy.

```
heights.cdf(68)      # 0.5
heights.cdf(72)      # 0.909
heights.sf(72)       # 0.091 = 1 - cdf,  $P(X > 72)$ 
```

8. Summary

Idea	Meaning	Heights $N(68, 3^2)$
Normal distribution	Symmetric bell-shaped continuous distribution	heights of adult men
Parameters	μ (centre), σ (spread)	68 and 3 inches
PDF	formula of Section 5	$f(72) = 0.0547$
CDF	area under the PDF up to x	$F(72) = 0.909$
Centre	mean = median = mode	68

- Other names: Gaussian distribution, bell curve.
- The tails approach the x axis but never touch it.
- The formula is a bell e^{-x^2} , shifted by μ , widened by σ , scaled to area 1.
- The CDF is an S-curve through 0.5 at the mean.

9. Sources

- Taylor, J. R. (1997). *An Introduction to Error Analysis*, 2nd ed. University Science Books. Chapter 5, “The Normal Distribution”.
- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 6.5 (Gaussian distribution).
- Rosenlicht, M. (1972). Integration in finite terms. *American Mathematical Monthly*, 79(9), 963-972. Shows that e^{-x^2} has no elementary antiderivative.

10. Key terms

Term	Meaning
Gaussian distribution $N(\mu, \sigma^2)$	Another name for the normal distribution A normal distribution with mean μ and variance σ^2
Tail (of a distribution)	The part of the curve far from the centre, where values are rare
Asymptotic	Coming ever closer to a line without touching it, like the normal curve’s tails and the x axis
Inflection point	A point where a curve switches between bending down and bending up; for the normal curve, at $\mu \pm \sigma$